BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

Name:

ID:

> **Due Date**
>
> Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\AllyK\bee 4750\hw5-ally,elliot\hw5-ally-elliot-1`
   Installed Measures        v0.3.3
   Installed GR_jll         v0.73.18+0
   Installed HiGHS_jll       v1.12.0+0
```

```
   Installed OpenBLAS32_jll     v0.3.29+0
   Installed PlotUtils        v1.4.4
   Installed OpenSSL         v1.6.0
   Installed MutableArithmetics   v1.6.7
   Installed Pango_jll        v1.57.0+0
   Installed FFMPEG          v0.4.5
   Installed StaticArraysCore    v1.4.4
   Installed DataStructures     v0.19.3
   Installed JSON           v1.3.0
   Installed GraphRecipes      v0.5.15
   Installed METIS_jll        v5.1.3+0
   Installed StatsBase        v0.34.8
   Installed Adapt          v4.4.0
   Installed ChainRulesCore     v1.26.0
   Installed HiGHS          v1.20.1
   Installed StableRNGs       v1.0.4
   Installed FFMPEG_jll       v8.0.0+0
   Installed Interpolations     v0.16.2
   Installed JuMP           v1.29.3
   Installed GR            v0.73.18
   Installed StructUtils       v2.6.0
   Installed ForwardDiff       v1.3.0
Precompiling project...
   1939.3 ms   Measures
   1408.5 ms   StableRNGs
   1421.1 ms   StaticArraysCore
   1649.7 ms   Adapt
   2460.4 ms   METIS_jll
   2463.0 ms   StructUtils
   1941.0 ms   EarCut_jll
   1978.2 ms   OpenBLAS32_jll
   4171.1 ms   ChainRulesCore
   2706.3 ms   ColorVectorSpace → SpecialFunctionsExt
   1374.3 ms   DiffResults
   5081.5 ms   DataStructures
   1459.8 ms   OffsetArrays → OffsetArraysAdaptExt
   1839.0 ms   Adapt → AdaptSparseArraysExt
   4050.0 ms   Pango_jll
   5050.8 ms   OpenSSL
   1750.5 ms   StructUtils → StructUtilsTablesExt
   5065.3 ms   FFMPEG_jll
   2435.9 ms   ChainRulesCore → ChainRulesCoreSparseArraysExt
   2600.0 ms   HiGHS_jll
```

```
  1549.4 ms    SortingAlgorithms
  4340.7 ms    LogExpFunctions → LogExpFunctionsChainRulesCoreExt
  5367.4 ms    SpecialFunctions → SpecialFunctionsChainRulesCoreExt
 11764.5 ms    ColorSchemes
  2751.7 ms    FFMPEG
 15558.1 ms    MutableArithmetics
 10077.5 ms    JSON
 10005.3 ms    ForwardDiff
  8005.2 ms    GR_jll
  6927.5 ms    StatsBase
  4707.2 ms    BenchmarkTools
 19787.7 ms    StaticArrays
  2486.4 ms    StaticArrays → StaticArraysStatisticsExt
  2578.9 ms    Adapt → AdaptStaticArraysExt
  2742.7 ms    StaticArrays → StaticArraysChainRulesCoreExt
  4003.1 ms    ArnoldiMethod
  3181.5 ms    ForwardDiff → ForwardDiffStaticArraysExt
  6187.1 ms    GeometryTypes
  6441.2 ms    Interpolations
  4638.7 ms    Interpolations → InterpolationsForwardDiffExt
 25918.7 ms    PlotUtils
 12542.7 ms    Graphs
  7924.9 ms    PlotThemes
 11468.5 ms    RecipesPipeline
 43609.3 ms    HTTP
 29958.5 ms    GeometryBasics
  4640.5 ms    NetworkLayout
 10811.8 ms    GR
  4907.5 ms    NetworkLayout → NetworkLayoutGraphsExt
  8911.2 ms    GraphRecipes
138232.4 ms    DataFrames
 22539.8 ms    Latexify → DataFramesExt
157227.9 ms    MathOptInterface
  6399.3 ms    MathOptIIS
 34198.7 ms    HiGHS
 70668.2 ms    JuMP
260191.9 ms    Plots
 13568.3 ms    Plots → GeometryBasicsExt
 58 dependencies successfully precompiled in 344 seconds. 194 already precompiled.
```

```julia
using JuMP
using HiGHS
```

```julia
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF**: Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF**: Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE**: Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using `JuMP` to solve the problem). Report the optimal objective value.

```
meow = Model(HiGHS.Optimizer)
# facilities in order: WTE, MRF, LF
S = [120, 90, 100]           # S_i  = waste production by city i (Mg/day)
K = [210, 350, 200]          # K_j  = capacity of facility j (Mg)
f = [0.1641, 1-0.3775]       # f_k  = fraction of waste at facility k left as residue (unitles
a = 1.5                      # a_ij = transportation cost from city i to facility j ($/Mg/km)
b = [60, 7 + 40 * f[2], 50]  # b_j  = variable cost of facility j ($/Mg)
c = [2500, 1500, 2000]       # c_j  = fixed cost of facility j ($/day)
l1 = [15 30 5                # l1_ij = distance from city i to facility j (km)
      10 25 15
      20 45 13]
l2 = [0 32 18                # l2_kj = distance from facility k to facility j (km)
      32 0 15
      18 15 0]
@variable(meow, W[1:3,1:3] >= 0)
@variable(meow, R[1:2,1:3] >= 0)
fix(R[1,1], 0, force=true)
fix(R[1,2], 0, force=true)
fix(R[2,2], 0, force=true)
@variable(meow, Y[1:3], Bin)
@constraint(meow, cities_dispose_all[i=1:3], sum(W[i,j] for j=1:3) == S[i])
@constraint(meow, facility_capacity[j=1:3], sum(W[i,j] for i=1:3) + sum(R[k,j] for k=1:2) <=
@constraint(meow, facility_status[j=1:3], !Y[j] => {sum(W[i,j] for i=1:3) == 0})
@constraint(meow, residual13, R[1,3] == f[1] * sum(W[i,1] for i=1:3))
@constraint(meow, residual21_23, R[2,1] + R[2,3] == f[2] * sum(W[i,2] for i=1:3))
@objective(meow, Min, sum(a * l1[i,j] * W[i,j] for i=1:3, j=1:3) + sum(a * l2[k,j] * R[k,j] :
optimize!(meow)
```

```
Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
```

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100$MW. In period 2, the demand is projected to be $d_2 = 1200$MW, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.